Multiplying and Dividing in Scientific Notation

Question Bank - Additional 4.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(3\times10^{2})(3\times10^{6})$

9×10^{8}

2. $\frac{6.3\times10^{7}}{2.1\times10^{3}}$

3×10^{4}

3. $(4\times10^{5})(8\times10^{6})$

$32\times10^{11}=3.2\times10^{12}$

4. $\frac{2.4\times10^{4}}{8\times10^{7}}$

$0.3\times10^{-3}=3\times10^{-4}$

5. $(2.5\times10^{-5})(6\times10^{-4})$

$15\times10^{-9}=1.5\times10^{-8}$

6. $\frac{9.9\times10^{-3}}{3.3\times10^{-8}}$

3×10^{5}

$-3-(-8)=5$.

Real quantities.

7. The Amazon River discharges about 5.5×10^{7} gallons of water every second.

a. About how many gallons does it discharge in one hour, which is 3.6×10^{3} seconds?

about 1.98×10^{11} ***gallons***

$5.5\cdot3.6=19.8$ *, so the product is* 19.8×10^{10} *before it is finished.*

b. A second river discharges about 1.1×10^{7} gallons per second. How many times larger is the Amazon's rate?

5 times larger

8. A factory produces about 6.4×10^5 bottles per day.

a. How many bottles does it produce in 3.5×10^2 days?

2.24×10^8 ***bottles***

$6.4 \cdot 3.5 = 22.4$.

b. A second factory produced 7.2×10^8 bottles in 1.2×10^3 days. How many bottles per day is that?

6×10^5 ***bottles per day***

Working backwards.

9. Find the missing number in each equation.

a. $(4 \times 10^n)(2 \times 10^3) = 8 \times 10^{10}$

n = 7

$n + 3 = 10$.

b. $\frac{7.2 \times 10^6}{n \times 10^2} = 2.4 \times 10^4$

n = 3

$7.2 \div n = 2.4$.

c. $(2.5 \times 10^3)(8 \times 10^5) =$ _____

2×10^9

The coefficients multiply to exactly 20.

Reading someone else's work.

10. Dana divides $\frac{4.8 \times 10^{11}}{1.2 \times 10^4}$ and writes 3.6×10^7 . Name her mistake and give the correct quotient.

She subtracted the coefficients instead of dividing them. The power of ten is right. The correct quotient is 4×10^7 ***.***

$4.8 - 1.2 = 3.6$ *, which is where her coefficient came from;* $4.8 \div 1.2 = 4$.

11. Which is the value of $\frac{9.6 \times 10^{-3}}{3.2 \times 10^5}$?

- ● **A.** 3×10^{-8}
- ○ B. 3×10^8
- ○ C. 6.4×10^{-8}
- ○ D. 3×10^2

A

12. Select ALL of the expressions whose value is 8×10^6 .

- ☒ **A.** $(1.6 \times 10^3)(5 \times 10^3)$
- ☒ **B.** $\frac{1.6 \times 10^{10}}{2 \times 10^3}$
- ☐ C. $(4 \times 10^2)(2 \times 10^8)$
- ☒ **D.** $\frac{4 \times 10^8}{5 \times 10^1}$
- ☒ **E.** $(4 \times 10^2)(2 \times 10^4)$
- ☐ F. $\frac{1.6 \times 10^{10}}{2 \times 10^4}$

A, B, D and E

B and D each give 0.8×10^7 *, which is* 8×10^6 *once it is finished.*